

Mid Term Portfolio
Option 2: Improving Tensorflow Model Performance and Quality

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Introduction

Neural networks can achieve strong predictive performance, but increasing the amount of training does not necessarily produce a better model. As training continues, a model may become increasingly optimized for the training data while providing little or no improvement when presented with unseen data. This behavior can contribute to overfitting and makes it important to evaluate model performance using validation and testing data rather than relying solely on improvements observed during training. Early stopping provides one approach for addressing this problem by monitoring model performance during training and terminating the process when additional epochs no longer produce meaningful improvement.

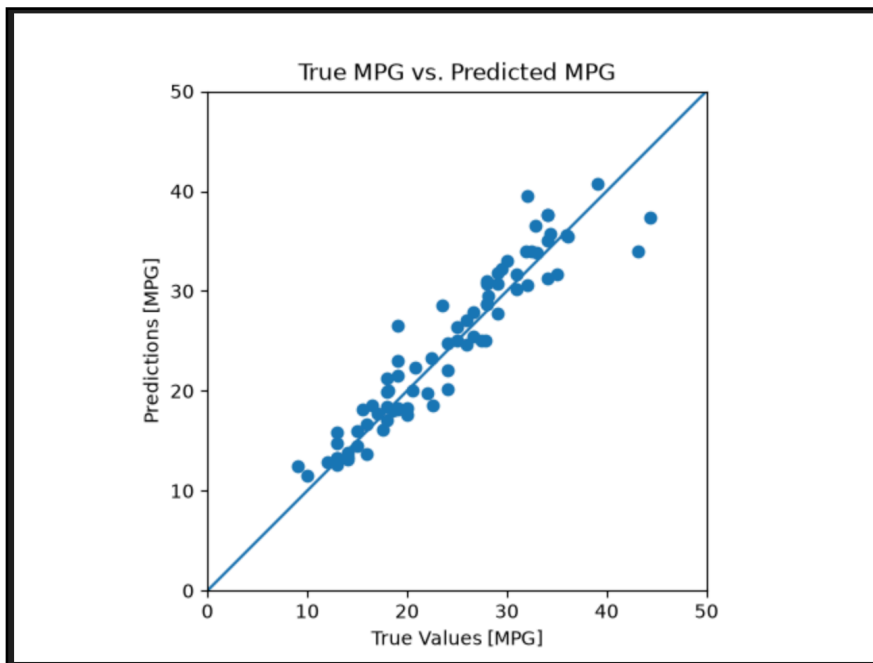
This project builds upon the TensorFlow regression model developed in Module 3 using the Auto MPG dataset. The original model was designed to predict vehicle fuel efficiency in miles per gallon (MPG) using vehicle characteristics such as cylinders, displacement, horsepower, weight, acceleration, model year, and origin. Although the original neural network continued training for 1,000 epochs, its validation performance stabilized considerably earlier while the training error continued to decline. The updated model therefore incorporates TensorFlow's `EarlyStopping` callback to reduce unnecessary training and limit model degradation. Model quality is then evaluated using mean absolute error (MAE), performance on previously unseen test data, predicted versus actual MPG values, and the distribution of prediction errors.

Although the Auto MPG dataset is not healthcare-related, the model-development concepts explored in this project are directly relevant to my work developing AuVentures, an AI-assisted platform intended to support longitudinal clinical reasoning for patients with complex conditions. A healthcare model that performs well on data it has already encountered but fails to generalize to new patient information would have limited practical value and could produce misleading results. Understanding how to recognize overfitting, establish stopping criteria, evaluate performance against unseen data, and examine the distribution of model errors therefore provides an important foundation for developing more reliable clinical AI systems. For AuVentures specifically, these techniques contribute to the broader goal of developing models that can identify meaningful patterns across longitudinal patient information while maintaining appropriate attention to model uncertainty, generalizability, and error.

Model Improvement Through Early Stopping

The original TensorFlow regression model developed in Module 3 was configured to train for 1,000 epochs using the RMSprop optimizer and mean squared error (MSE) as the loss function. As shown in Figure 1, the training mean absolute error (MAE) continued to decrease throughout the training process, eventually approaching approximately 1 MPG. However, the validation MAE declined rapidly during the initial epochs and then stabilized at approximately 2.3 MPG. The growing difference between training and validation error indicates that additional training was improving the model's fit to the training data without producing a comparable improvement on validation data. In this case, completing all 1,000 epochs provided little additional value and increased the potential for overfitting.

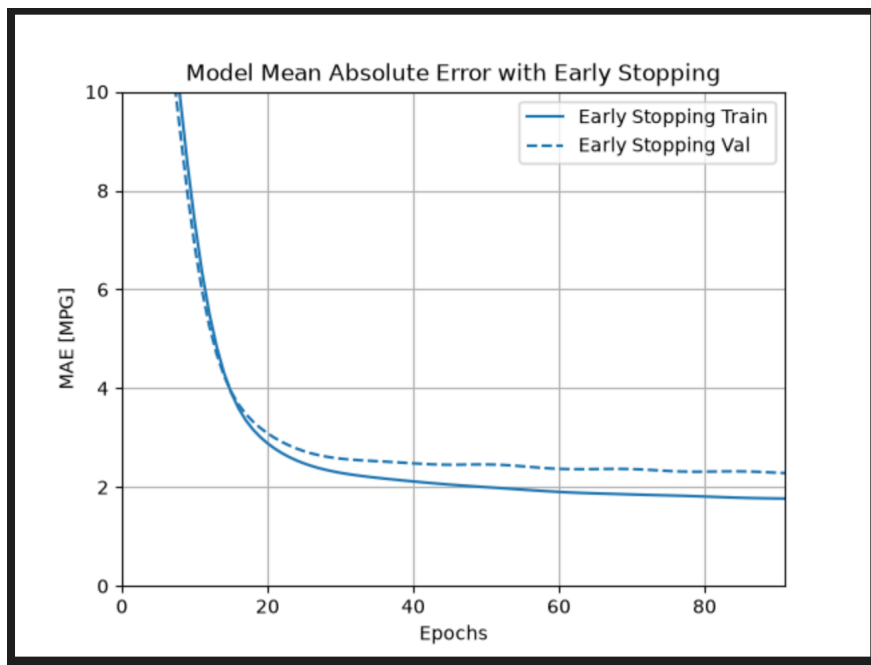
Figure 1. *Original model training and validation MAE over 1,000 epochs.*



To address this behavior, the updated model incorporated TensorFlow's `EarlyStopping` callback with validation loss (`val_loss`) as the monitored metric and a patience value of 10 epochs. This configuration allowed the model to continue training while validation performance was improving but automatically stopped training after 10 consecutive epochs without sufficient improvement. The model was also configured with `restore_best_weights=True`, allowing it to return to the weights associated with its best validation-loss performance rather than retaining the weights from the final epoch.

As shown in Figure 2, `EarlyStopping` terminated training after only 92 epochs, substantially reducing the amount of training compared with the original 1,000-epoch model. At the end of the recorded training history, training MAE was approximately 1.77 MPG and validation MAE was approximately 2.24 MPG. More importantly, the validation curve had already begun to stabilize while the training curve continued to decrease. Early stopping therefore reduced unnecessary training while preserving approximately the same level of validation performance observed in the much longer Module 3 run. The result demonstrates that additional epochs are not necessarily evidence of model improvement; validation performance provides a more meaningful indication of when continued training is no longer improving the model's ability to generalize.

Figure 2. *Training and validation MAE after implementing `EarlyStopping`; training automatically terminated after 92 epochs.*



Test Set Evaluation and Model Generalization

After training was completed, the model was evaluated using the test dataset that had been withheld from the training process. This step is important because training and validation performance alone do not demonstrate how well a model will perform when presented with previously unseen observations. The test set provides a closer approximation of how the model may perform when applied to new data. This distinction is particularly important for predictive models intended for real-world applications, where the objective is not simply to reproduce patterns within training data but to generalize those patterns to new observations.

The EarlyStopping model achieved a test mean absolute error (MAE) of 2.04 MPG and a mean squared error (MSE) of 7.19, as shown in Figure 3. An MAE of 2.04 MPG means that, on average, the model's predictions differed from the actual fuel-efficiency values by approximately 2 MPG. This result is also slightly better than the validation MAE observed near the end of training, which was approximately 2.24 MPG. Because the test data were not used to train the neural network, the relatively similar validation and test errors provide evidence that the model is generalizing reasonably well to unseen data rather than achieving its performance primarily through fitting the training observations.

Figure 3. *Evaluation of the EarlyStopping model on the previously unseen test dataset.*

```
.....  
Training stopped after 92 epochs.
```

```
Training History:
```

	loss	mae	mse	val_loss	val_mae	val_mse	epoch
87	6.559223	1.781504	6.559223	9.100489	2.308681	9.100489	87
88	6.457080	1.774230	6.457080	9.226078	2.315443	9.226078	88
89	6.427528	1.749849	6.427528	9.142832	2.218932	9.142832	89
90	6.457749	1.767802	6.457749	9.104190	2.284149	9.104190	90
91	6.314684	1.765581	6.314684	9.158989	2.240744	9.158989	91

```
3/3 - 0s - 4ms/step - loss: 7.1908 - mae: 2.0365 - mse: 7.1908
```

```
Testing Set Mean Abs Error: 2.04 MPG
```

```
Testing Set Mean Squared Error: 7.19
```

```
Sample Test Predictions:
```

```
True MPG: 15.0 | Predicted MPG: 16.0  
True MPG: 10.0 | Predicted MPG: 11.5  
True MPG: 9.0 | Predicted MPG: 12.5  
True MPG: 25.0 | Predicted MPG: 25.1  
True MPG: 19.0 | Predicted MPG: 21.6  
True MPG: 14.0 | Predicted MPG: 13.4  
True MPG: 14.0 | Predicted MPG: 13.2  
True MPG: 13.0 | Predicted MPG: 13.3  
True MPG: 18.0 | Predicted MPG: 20.0  
True MPG: 35.0 | Predicted MPG: 31.7
```

```
Prediction Error Statistics:
```

```
count    78.000000  
mean      0.493949  
std       2.652748  
min      -9.169294  
25%     -0.879694  
50%      0.768072  
75%      1.961959  
max       7.596949
```

```
Name: MPG, dtype: float64
```

```
Process finished with exit code 0
```

```
> assignments > Portfolio_Project > tensorflow_model_performance.py
```

The test results also reinforce the value of stopping training when validation performance no longer improves. The original model continued for 1,000 epochs, whereas the updated model stopped after 92 epochs and still achieved an average test error of approximately 2 MPG. The objective of EarlyStopping was therefore not necessarily to produce a dramatically lower error than the original model, but to prevent hundreds of additional epochs from continuing to optimize the training data without evidence of corresponding improvement in generalization. In this case, the results suggest that a substantially shorter training process was sufficient to produce a model capable of making reasonably accurate predictions on unseen vehicle data.

This concept has particular relevance to future healthcare applications. A clinical model cannot be evaluated solely by how accurately it represents the patient information on which it was developed. Its practical value depends on whether learned patterns remain useful when the model encounters new patients, new observations, or future longitudinal data. The test-set evaluation in this experiment therefore represents a simplified example of a broader requirement for the

AuVentures work: model performance must ultimately be evaluated on information that was not used to train the model before its predictions can be considered meaningfully generalizable.

Conclusion

This project demonstrated that improving a neural network does not necessarily require increasing its complexity or extending its training time. The original TensorFlow regression model was allowed to train for 1,000 epochs, even though validation performance stabilized much earlier. By incorporating an `EarlyStopping` callback, the updated model automatically terminated training after 92 epochs when validation loss stopped improving. Despite using substantially fewer training epochs, the model achieved a test MAE of 2.04 MPG and an MSE of 7.19, demonstrating that the additional epochs used by the original model were not necessary to achieve reasonable performance on unseen data.

The prediction results provided additional evidence of the model's ability to generalize. Predicted MPG values generally followed the actual MPG values, with most observations appearing relatively close to the ideal prediction line. Analysis of the prediction errors also showed that errors were concentrated near zero, although several larger positive and negative errors remained. These results demonstrate why model quality should be evaluated using multiple measures rather than training accuracy or loss alone. Validation behavior, test-set performance, individual predictions, and error distributions each provide different information about the strengths and limitations of a trained model.

The lessons from this exercise are also applicable to the longer-term development of AuVentures. Although predicting vehicle MPG is considerably different from analyzing longitudinal healthcare information, both applications require models that can identify patterns without becoming overly dependent on the data used during training. In a healthcare environment, the consequences of poor generalization are also considerably more significant, making techniques for controlling overfitting and evaluating unseen data especially important. This exercise reinforces that developing reliable AI is not simply a matter of achieving the lowest possible training error. It requires understanding when a model has learned enough, recognizing where errors remain, and evaluating whether its learned patterns continue to perform when presented with new information. These principles will be important as AuVentures progresses toward AI-assisted longitudinal pattern recognition and clinical reasoning support.

References

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